413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 13

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Recall

In Lecture 12, we considered the linear regression model

$$\mathbf{y} = X\beta + \varepsilon,$$

with

$$\mathbb{E}[\varepsilon \mid X] = 0, \qquad \mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

Ordinary least squares gives

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.}$$

We also showed that

$$\boxed{\mathbb{E}[\hat{\beta} \mid X] = \beta}$$

and

$$\boxed{\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X^T X)^{-1}.}$$

We now ask whether the fitted model provides an adequate description of the observed data and whether the assumptions used to derive these results are reasonable.

1 Assessing Model Fit

Suppose the observed responses are

$$y_1, \ldots, y_n.$$

If we ignore the predictors, the squared error obtained by predicting every response using the sample mean $\bar{y}$ is

$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2,$$

called the *total sum of squares*. After fitting the regression model, the fitted values are

$$\hat{y}_i = \mathbf{x}_i^T \hat{\beta},$$

and the fitted residuals are

$$e_i = y_i - \hat{y}_i.$$

The residual sum of squares is

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} e_i^2.$$

Thus, TSS measures the squared error of the mean-only predictor, while RSS measures the squared error after including the predictors.

1.1 Coefficient of Determination

For ordinary least squares with an intercept,

$$RSS \leq TSS.$$

We define the coefficient of determination

$$R^2 = 1 - \frac{RSS}{TSS}.$$

Equivalently,

$$R^2 = \frac{TSS - RSS}{TSS}.$$

Therefore,

$$R^2 = \textbf{fraction of the mean-only squared error removed by the regression model}.$$

If

$$R^2 = 0,$$

the fitted model does no better, in squared-error terms, than predicting $\bar{y}$. If

$$R^2 = 1,$$

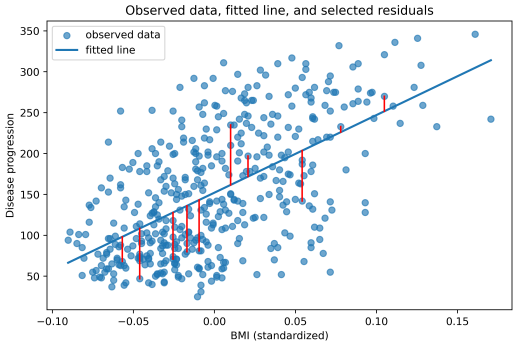
then

$$RSS = 0,$$

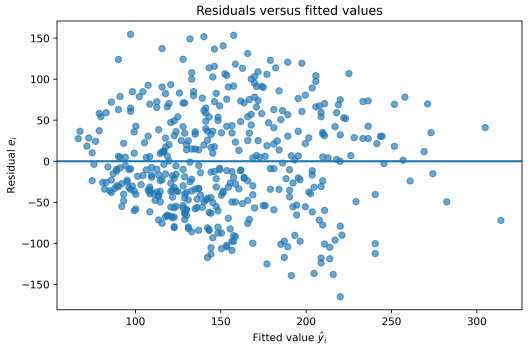
and every observed response lies exactly on the fitted regression model.

When predictors are added to an ordinary least-squares model, the training RSS cannot increase. Consequently, the training R^2 cannot decrease.

Larger training R^2 does not by itself imply better prediction on new data.



(a) Observed data and fitted regression line. The vertical distance between an observation y_i and its fitted value $\hat{y}_i$ is the residual $e_i = y_i - \hat{y}_i$.



(b) Residuals versus fitted values. For an adequately specified linear model, the residuals should be centered around zero without systematic structure.

Figure 1: Regression fit and corresponding residual diagnostic. The residual plot makes systematic departures from the fitted mean structure easier to detect.

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2 Regression Assumptions and Residual Diagnostics

For the linear model

$$\mathbf{y} = X\beta + \varepsilon,$$

the assumptions used in the previous lecture can be written as

$$\mathbb{E}[\varepsilon \mid X] = 0,$$

$$\mathrm{Var}(\varepsilon_i \mid X) = \sigma^2,$$

$$\mathrm{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \qquad i \neq j.$$

For exact Gaussian likelihood-based inference, we additionally assume

$$\varepsilon_i \mid X \sim \mathcal{N}(0, \sigma^2).$$

The errors ε_i are unobserved. After fitting the model, we examine the residuals

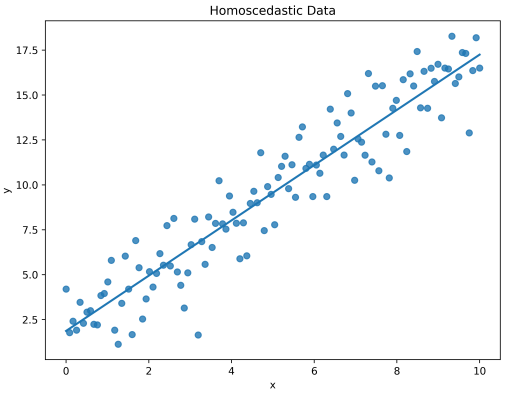
$$e_i = y_i - \hat{y}_i$$

as estimates of the errors.

A residual plot displays

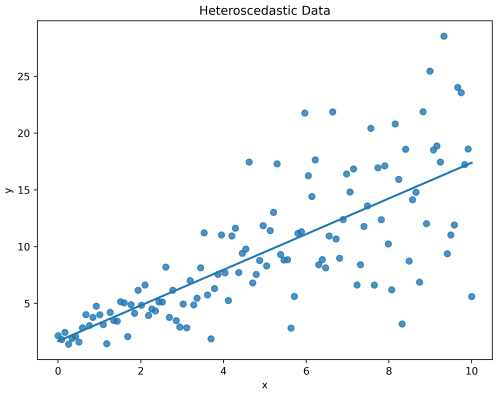
$$e_i \quad \text{versus} \quad \hat{y}_i$$

or versus a predictor x_i. For an adequately specified linear model, the residuals should show no systematic structure around zero. Systematic patterns can indicate violations of the model assumptions.



(a) Homoscedastic errors. The conditional variance is constant:

$$\mathrm{Var}(Y \mid X = x) = \sigma^2.$$



(b) Heteroscedastic errors. The conditional variance changes with x:

$$\mathrm{Var}(Y \mid X = x) = \sigma^2(x).$$

Figure 2: Comparison of homoscedastic and heteroscedastic regression models. The conditional mean may be the same in both cases, while the conditional variance behaves differently.

3 Homoscedasticity and Heteroscedasticity

The homoscedastic regression model assumes

$$\boxed{\mathrm{Var}(\varepsilon_i \mid X) = \sigma^2 \qquad \text{for every } i.}$$

Thus, the conditional spread of Y around the regression mean is the same across predictor values.

In a heteroscedastic model,

$$\mathrm{Var}(\varepsilon_i \mid X) = \sigma_i^2,$$

where the error variance may vary between observations. In matrix form,

$$\mathrm{Var}(\varepsilon \mid X) = \Omega,$$

rather than

$$\sigma^2 I_n.$$

The OLS estimator is still

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.$$

If

$$\mathbb{E}[\varepsilon \mid X] = 0,$$

it remains unbiased:

$$\mathbb{E}[\hat{\beta} \mid X] = \beta.$$

However, its covariance is now

$$\mathrm{Var}(\hat{\beta} \mid X) = (X^T X)^{-1} X^T \Omega X (X^T X)^{-1},$$

rather than

$$\sigma^2 (X^T X)^{-1}.$$

Therefore, heteroscedasticity primarily affects the estimated uncertainty of the regression coefficients.

4 Outliers, Leverage, and Influence

An unusual observation can affect a regression model in different ways. It is important to distinguish three concepts.

4.1 Outlier

An observation has a large residual if

$$\boxed{|e_i| = |y_i - \hat{y}_i|}$$

is large. Such an observation has an unusual response value relative to the fitted model at its predictor value.

4.2 Leverage

An observation has high leverage when its predictor values are unusual relative to the other observations. For linear regression, define the hat matrix

$$H = X(X^T X)^{-1} X^T.$$

Since

$$\hat{\mathbf{y}} = X\hat{\beta} = H\mathbf{y},$$

the diagonal element

$$\boxed{h_{ii} = H_{ii}}$$

measures the leverage of observation i. For simple linear regression with an intercept,

$$\boxed{h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{n} (x_j - \bar{x})^2}.}$$

Thus, observations far from $\bar{x}$ have larger leverage.

4.3 Influence

An observation is influential if removing it substantially changes the fitted regression model. High leverage alone does not imply high influence. For example, an observation can have an extreme predictor value but lie close to the existing regression trend. Such a point may have high leverage but only a small effect on the fitted coefficients. A point is often influential when it has both **high leverage** and **a substantial residual**. Thus,

$$\text{outlier} \neq \text{high leverage} \neq \text{influential observation.}$$

4.4 Why High-Leverage Points Can Strongly Affect the Slope

For simple linear regression,

$$\hat{\beta}_1 = \frac{\sum_{j=1}^{n}(x_j - \bar{x})(y_j - \bar{y})}{\sum_{j=1}^{n}(x_j - \bar{x})^2}.$$

The sensitivity of the fitted slope to the response y_i is

$$\boxed{\frac{\partial \hat{\beta}_1}{\partial y_i} = \frac{x_i - \bar{x}}{\sum_{j=1}^{n}(x_j - \bar{x})^2}.}$$

Therefore, observations with predictor values far from $\bar{x}$ have greater leverage: a change in their response value can produce a larger change in the fitted slope.

5 Collinearity

Consider a multiple linear regression model with predictors X_1 and X_2:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon.$$

Suppose the two predictors are nearly linearly dependent. For example,

$$X_2 \approx cX_1.$$

Then the two predictor columns contain nearly redundant information. The OLS estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.$$

If the columns of X are nearly linearly dependent, then $X^T X$ is nearly singular. From

$$\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X^T X)^{-1},$$

we see that near-singularity of $X^T X$ can produce large variances in the estimated coefficients. Thus,

$$\boxed{\text{strong collinearity} \Longrightarrow \text{unstable individual coefficient estimates.}}$$

Importantly, the fitted predictions

$$X\hat{\beta}$$

can remain relatively stable even when individual coefficients vary substantially.

5.1 Conditioning of the Design Matrix

For a symmetric positive-definite matrix $X^T X$, define the condition number

$$\kappa(X^T X) = \frac{\lambda_{\max}(X^T X)}{\lambda_{\min}(X^T X)}.$$

If the columns of X become nearly linearly dependent, then

$$\lambda_{\min}(X^T X) \to 0,$$

and therefore

$$\kappa(X^T X)$$

becomes large. A large condition number indicates that small perturbations in the data may produce large changes in the estimated coefficients. If the columns of X are exactly linearly dependent, then

$$X^T X$$

is singular and the ordinary inverse

$$(X^T X)^{-1}$$

does not exist.

6 Assessing Gaussian Errors with a Q–Q Plot

Suppose the regression model assumes

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

After fitting the model, we can assess this assumption using the residuals

$$e_1, \ldots, e_n.$$

First, order the residuals:

$$e_{(1)} \leq e_{(2)} \leq \cdots \leq e_{(n)}.$$

Associate the ith ordered residual with the probability

$$p_i = \frac{i - \frac{1}{2}}{n}.$$

Let

$$\Phi(z) = P(Z \leq z), \qquad Z \sim \mathcal{N}(0, 1),$$

denote the standard normal cumulative distribution function. The corresponding theoretical Gaussian quantile is

$$z_i = \Phi^{-1}(p_i).$$

A normal Q-Q plot displays the pairs

$$(z_i, e_{(i)}), \qquad i = 1, \ldots, n.$$

6.1 Why Gaussian Residuals Produce a Straight Line

If

$$E \sim \mathcal{N}(\mu, \sigma^2),$$

then its quantile function satisfies

$$Q_E(p) = \mu + \sigma \Phi^{-1}(p).$$

Therefore,

$$e_{(i)} \approx \mu + \sigma z_i.$$

Hence, if the residual distribution is approximately Gaussian, the Q-Q plot should be approximately linear. For regression residuals, we expect

$$\mu \approx 0.$$

Systematic departures from a straight line indicate departures from Gaussianity.

A Q-Q plot assesses the *distributional shape* of the residuals. It does not test whether the residual variance is constant across predictor values. Therefore,

$$\boxed{\text{Q-Q plot} \rightarrow \text{Gaussian-error assumption,}}$$

while

$$\boxed{\text{residual-versus-fitted plot} \rightarrow \text{mean structure and variance structure.}}$$

7 Coefficient Uncertainty

From Lecture 12,

$$\text{Var}(\hat{\beta} \mid X) = \sigma^2 (X^T X)^{-1}.$$

Therefore, for coefficient $\hat{\beta}_j$,

$$\text{Var}(\hat{\beta}_j \mid X) = \sigma^2 \left[(X^T X)^{-1}\right]_{jj}.$$

Replacing the unknown σ by an estimate $\hat{\sigma}$ gives the standard error

$$SE(\hat{\beta}_j) = \hat{\sigma} \sqrt{\left[(X^T X)^{-1}\right]_{jj}}.$$

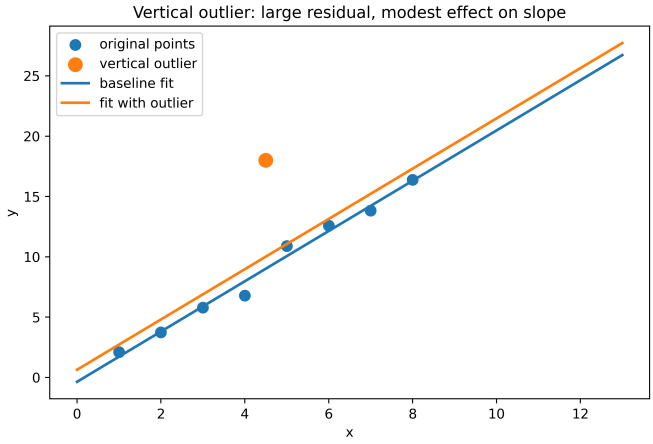
Thus, reporting only the fitted coefficient $\hat{\beta}_j$ is incomplete. Its magnitude should be interpreted together with its sampling uncertainty.

Summary

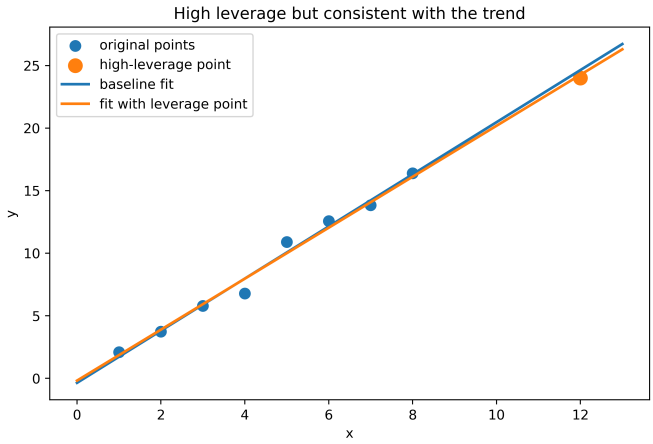
After fitting a regression model, we should ask several distinct questions:

R^2	How much squared error is removed relative to the mean-only model?
Residual plot	Is the conditional mean and variance structure reasonable?
Q–Q plot	Are the residuals approximately Gaussian?
Leverage/influence	Are individual observations strongly affecting the fit?
Conditioning	Are the regression coefficients stable under correlated predictors?

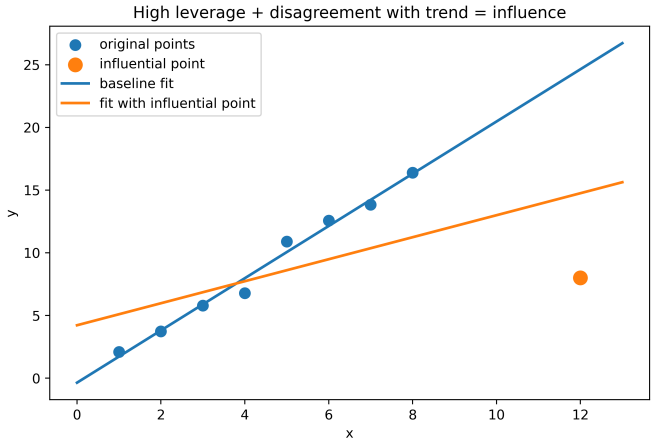
Fitting the coefficients is therefore only the first step. A regression analysis also requires checking the assumptions under which those coefficients and their uncertainties are interpreted.



(a) Vertical outlier. The point has a large residual, but its predictor value lies within the range of the remaining data. Its effect on the fitted slope is therefore modest.



(b) High-leverage point consistent with the trend. The predictor value is far from the center of the observed x values, but the point is consistent with the existing linear trend. It has high leverage but little influence on the fitted line.



(c) Influential point. The point has both high leverage and substantial disagreement with the existing trend, producing a large change in the fitted regression line.

Figure 3: Distinction between a vertical outlier, a high-leverage observation, and an influential observation. High leverage alone does not imply high influence; influence arises when an observation has sufficient leverage and substantially disagrees with the fitted trend.